



Assiut
University

Faculty of Computers and
Information
Assiut University
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Course: Computer Graphics
Code: CS351



Question Bank أسئلة إسترشادية

True or False?

1. The volume of a small cube change when its vertices are rotated. **False**
2. The projection preserves straight lines. **True**
3. The dot product of any two perpendicular vectors equals 0 **True**
4. With color tracking, you can tell OpenGL to set material properties by only calling glColor. **True**
5. $\text{scale}(z, 3) * \text{scale}(z, 4) = \text{scale}(z, 4) * \text{scale}(z, 3)$ **True**

Choose the correct answer for the following questions:

6. What is the correct order of operations of the OpenGL graphics pipeline?
a) **modelview transformation, projection transformation, divide by w, viewport transform**
b) projection transformation, modelview transformation, divide by w, viewport transform
c) modelview transformation, divide by w, projection transformation, viewport transform
d) modelview transformation, viewport transform, divide by w, projection transformation
7. Given $U = (1, 2, -1)$ and $V = (0, -1, 3)$, the cross-product $U \times V$ equals
.....
a) (0,-2,-3)
b) **(5,-3,-1)**
c) (1, 1,2)
d) (5,3,-2)
8. The process of backface..... makes rendering objects quicker and more efficient by reducing the number of polygons for the program to draw.
a) FOV
b) **Culling**
c) Antialiasing
d) Mipmapping

9. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef (-1.0,0.5, -1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0.5 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- a) A
b) B
c) C
d) D

10. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(-90.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- a) A
b) B
c) C
d) D

11. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(-90.0, 0, 0, 1.0);
glRotatef(90.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- a) A
b) B
c) C

d) **D**

12. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glScalef(-1.0, 2.0, -1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

a) A

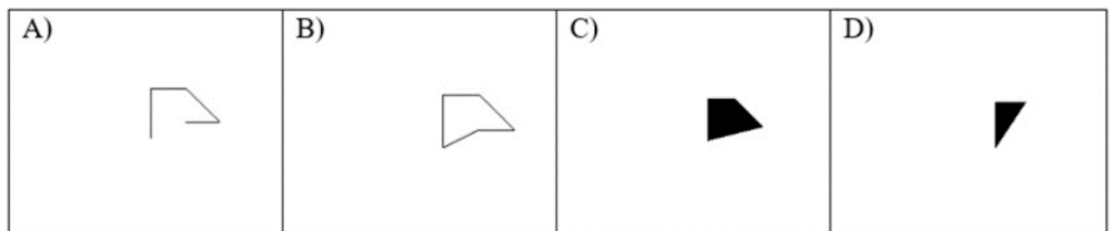
b) **B**

c) C

d) D

13. Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);  
glVertex2f(2.0, 0.0);  
glVertex2f(2.0, 3.0);  
glVertex2f(4.0, 3.0);  
glVertex2f(6.0, 1.0);  
glVertex2f(4.0, 1.0);  
glEnd();
```



a) **A**

b) B

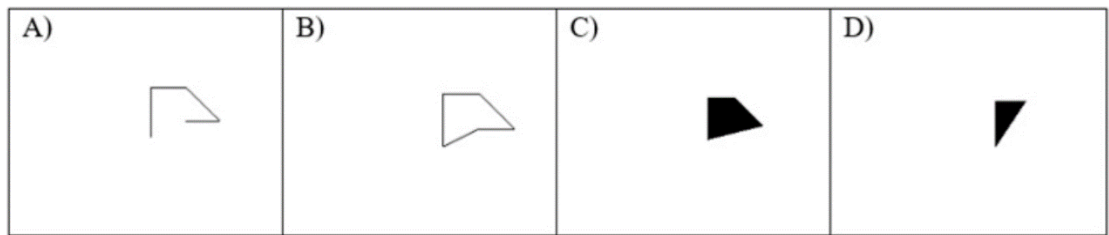
c) C

d) D

14. Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);  
glVertex2f(2.0, 0.0);  
glVertex2f(2.0, 3.0);  
glVertex2f(4.0, 3.0);  
glVertex2f(6.0, 1.0);  
glVertex2f(4.0, 1.0);
```

glEnd();



- a) A
- b) B**
- c) C
- d) D

15. Select the output of executing the following OpenGL code segment:

```
glBegin(GL_QUADS);  
  glVertex2f(2.0, 0.0);  
  glVertex2f(2.0, 3.0);  
  glVertex2f(4.0, 3.0);  
  glVertex2f(6.0, 1.0);  
  glVertex2f(4.0, 1.0);  
glEnd();
```

- a) A
- b) B
- c) C**
- d) D

16. The OpenGL function used to set up a parallel view volume is

- a) glutCreateWindow
- b) glRectf
- c) glOrtho**
- d) glViewport

17. The following two code segments are equivalent:

Code#1

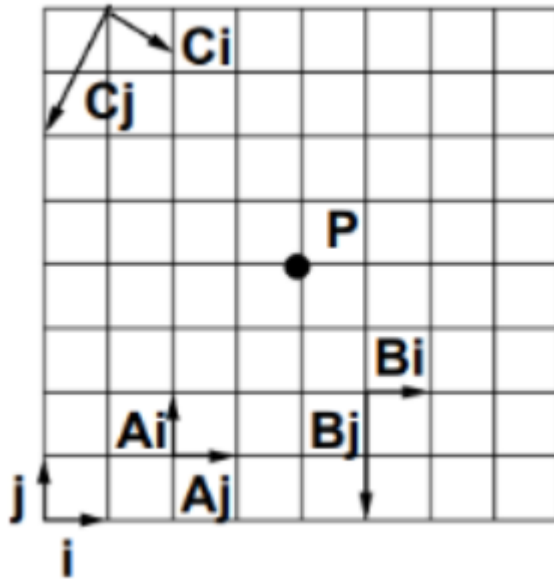
```
glScalef(1.0, 1.0, 2.0);  
glScalef(1.0, 1.0, 4.0);  
dawShape_3();
```

Code#2

```
glScalef(1.0, 1.0, 4.0);  
glScalef(1.0, 1.0, 2.0);  
dawShape_3();
```

- a) True**
- b) False

The figure contains a point coordinate P that can be expressed as $P = 4*i + 4*j$, where i and j are basis vectors of unit length along the x and y axes, respectively. Using the figure below to answer the following questions.

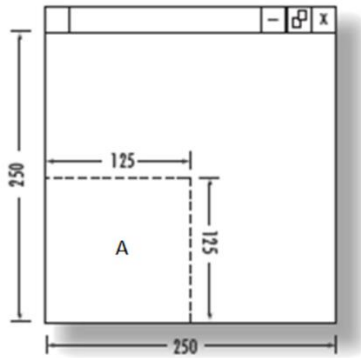


18. is the coordinates of point P with respect to coordinate frames A
 - a) (3,2)
 - b) (2,3)
 - c) (4,4)
 - d) (-2,3)

19. is the coordinates of point P with respect to coordinate frames B
 - a) (1,2)
 - b) (-2,-2)
 - c) (2,1)
 - d) (-1,-1)

20. The OpenGL command `glClearColor(0.0f, 0.0f, 1.0f, 1.0f)` clear a window with blue color
 - a) True
 - b) False

21. To map the output of the the OpenGL code to the area A (as shown in the figure), we use the .



a) `glViewport(0, 0, 125, 125);`

b) `glViewport(125, 125, 125, 125);`

c) `glViewport(125, 125, 0, 0);`

d) `glViewport(0, 125, 0, 125);`

22. Given three vertices $P_1=(1, 1, 0.5)$, $P_2=(1.5, 0.5, 0.5)$, $P_3=(1, 0.5, 1)$ on a plane, the equation of the plane is

a) $x+y+0.5z=2.25$

b) $x+2y+2z=3.5$

c) $x+2y+z=3$

d) $x+y+z=2.5$

23.

```
#define X .525731112119133606
```

```
#define Z .850650808352039932
```

```
static GLfloat vdata[12][3] = {
    {-X, 0.0, Z}, {X, 0.0, Z}, {-X, 0.0, -Z}, {X, 0.0, -Z},
    {0.0, Z, X}, {0.0, Z, -X}, {0.0, -Z, X}, {0.0, -Z, -X},
    {Z, X, 0.0}, {-Z, X, 0.0}, {Z, -X, 0.0}, {-Z, -X, 0.0}
};
```

```
static GLint tindices[20][3] = {
    {0,4,1}, {0,9,4}, {9,5,4}, {4,5,8}, {4,8,1},
    {8,10,1}, {8,3,10}, {5,3,8}, {5,2,3}, {2,7,3},
    {7,10,3}, {7,6,10}, {7,11,6}, {11,0,6}, {0,1,6},
    {6,1,10}, {9,0,11}, {9,11,2}, {9,2,5}, {7,2,11} };
```

The OpenGL code to render the icosahedron is

A)

```
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
        glVertex3fv(&vdata[tindices[i][0]][0]);
        glVertex3fv(&vdata[tindices[i][1]][0]);
        glVertex3fv(&vdata[tindices[i][2]][0]);
    glEnd();
}
```

B)

```
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
        glVertex3fv(&tindices[vdata[i][0]][0]);
        glVertex3fv(&tindices[vdata[i][1]][0]);
        glVertex3fv(&tindices[vdata[i][2]][0]);
    glEnd();
}
```

C)

```
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
        glVertex3fv(tindices[&vdata[i][0]][0]);
        glVertex3fv(tindices[&vdata[i][1]][0]);
        glVertex3fv(tindices[&vdata[i][2]][0]);
    glEnd();
}
```

D)

```
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
        glVertex3fv(vdata[&tindices[i][0]][0]);
        glVertex3fv(vdata[&tindices[i][1]][0]);
        glVertex3fv(vdata[&tindices[i][2]][0]);
    glEnd();
}
```

- a) A
- b) B
- c) C
- d) D

24. The OpenGL command(s) that makes the transformation of Figure A to Figure B:

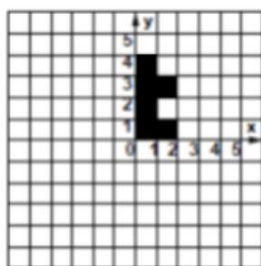


Figure: A

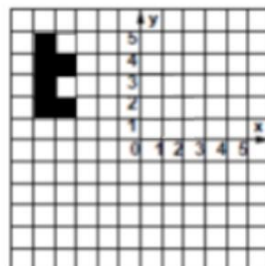


Figure: B

- a) `glTranslatef(-5.0, 0.0, 1.0);`
- b) `glTranslatef(-5.0, 1.0, 0.0);`
- c) `glRotatef(90.0, 0.0, 0.0, 1.0); glTranslatef(-1.0, 1.0, 0.0);`
- d) `glTranslatef(-1.0, 1.0, 0.0); glRotatef(90.0, 0.0, 0.0, 1.0);`

25. The normal vector for the plane $2x+2y+z=1$ is:
- a) $(1,1,1/3)$
 - b) $(0,1,0)$
 - c) $(1/3,2/3,1/3)$
 - d) $(2/3,2/3,1/3)$
26. $\text{Translate}(1,2,3) * \text{Rotate}(90.0, 0, 0, 1.0) = \text{Rotate}(90.0, 0, 0, 1.0) * \dots\dots\dots$
- a) $\text{Translate}(2,-1,3)$
 - b) $\text{Translate}(1,2,3)$
 - c) $\text{Translate}(1,3,-2)$
 - d) $\text{Translate}(-2,1,3)$
27. $\dots\dots\dots$ light component comes from one direction and once it hits a surface, it is scattered.
- a) Diffuse
 - b) Specular
 - c) Ambient
 - d) Spot
28. $\dots\dots\dots$ light comes from one direction and tends to bounce off the surface in a preferred direction.
- a) Diffuse
 - b) Specular
 - c) Ambient
 - d) Spot
29. $\dots\dots\dots$ light is so scattered; it is impossible to tell what its original source was.
- a) Diffuse
 - b) Specular
 - c) Ambient
 - d) Spot
30. $\dots\dots\dots$ projection preserves parallel lines.
- a) Orthographic
 - b) Perspective
 - c) Frustum
 - d) Symmetric

31. The dot product of any two perpendicular vectors equals

- a) 0
- b) 1
- c) (0,0,0)
- d) (1,1,1)

32. If the GL_SPOT_CUTOFF is, spotlight effect equals one.

- a) 0
- b) 45
- c) 90
- d) 180

33. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef (-1.0,0.5, -1.0);
glRotatef(-90.0, 0, 0, 1.0);
glPopMatrix();
glRotatef(90.0, 0, 0, 1.0);
glScalef(-1.0, 2.0,-1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

34. The RGB of A cyan color is

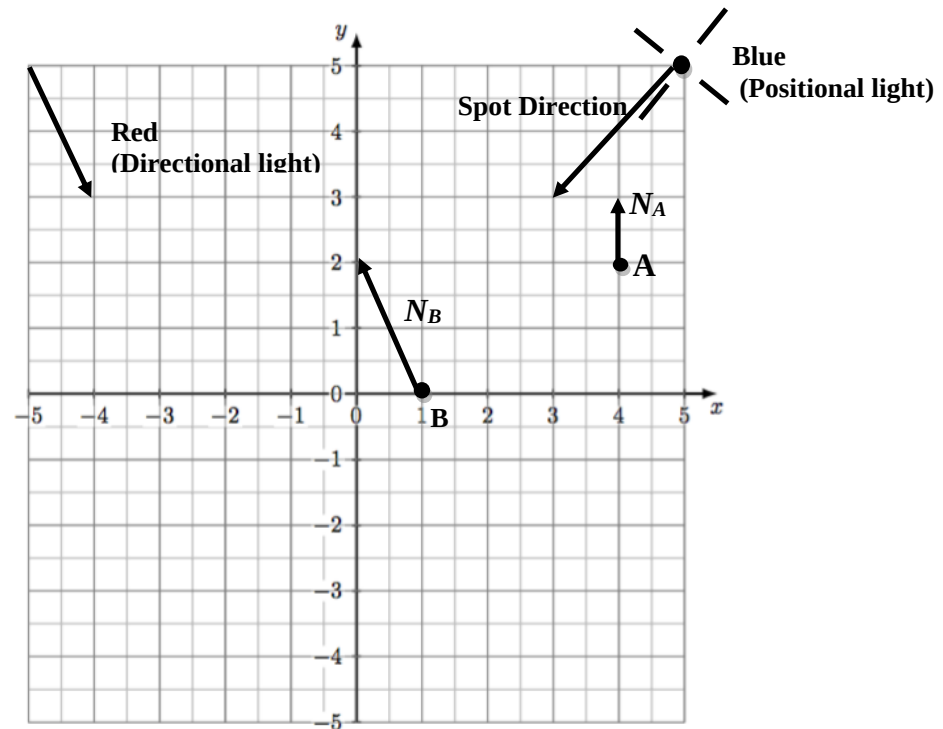
- a) (1.0,1.0,0.0)
- b) (1.0,0.0,1.0)
- c) (0.0,1.0,1.0)
- d) (1.0,1.0,1.0)

35. Convert the following Target Camera to Free Camera

Center = (1, -1, 0)
 Eye = (2, 2, 2)
 Up = (0, 1, 0)

A)	B)	C)	D)
Center = (3, 1, 2) Eye = (2, 2, 2) Up = (0, 1, 0)	Foward = (-1, -3, -2) Eye = (2, 2, 2) Up = (0, 1, 0)	Center = (1, -1, 0) Foward = (-1, -3, -2) Up = (0, 1, 0)	Center = (1, -1, 0) Foward = (-2, -3, -1) Up = (0, 1, 0)

The figure below contains two light sources and two points in 2D. The left light source emits red diffuse light = (1.0, 0.1, 0.1) and its spotlight effect=1. The right light emits blue diffuse light = (0.1, 0.2, 1.0), spot cutoff = 20, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =2.0, quadratic attenuation=0, the material ambient = (0, 0, 0), the material emission=(0.1,0.1,0.1), the diffuse components of the material = (0.6,0.8,0.4), and the specular components of the material = (0, 0, 0) for the two points. Using the Phong reflection model, answer the following questions:



36. The normalized normal vector N_A =

A) (0,1)	B) (1,0)	C) (0,-2)	D) (-2,0)
----------	----------	-----------	-----------

37. The normalized normal vector N_B =

A) $\left(\frac{-1}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$	B) $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$	C) $\left(\frac{-2}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	D) $\left(\frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$
---	---	---	---

38. The diffuse intensity for the point A for the blue light source =

A) $\frac{1}{2\sqrt{10}}$	B) $\frac{3}{\sqrt{5}}$	C) $\frac{2}{\sqrt{5}}$	D) $\frac{3}{\sqrt{10}}$
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39. The attenuation factor for the point A for the blue light source =

A) $\frac{1}{\sqrt{10}}$	B) 1	C) $\frac{1}{2\sqrt{41}}$	D) $\frac{1}{2\sqrt{10}}$
--------------------------	------	---------------------------	---------------------------

40. The attenuation factor for the point B for the red light source =

A) $\frac{1}{\sqrt{10}}$	B) $\frac{1}{2\sqrt{41}}$	C) 1	D) $\frac{1}{2\sqrt{10}}$
--------------------------	---------------------------	------	---------------------------

41. The spotlight effect for the blue light for point A =

A) 0	B) 1	C) $\frac{81}{82}$	D) $\frac{82}{81}$
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42. The spotlight effect for the blue light for point B =

A) 0	B) 1	C) $\frac{81}{82}$	D) $\frac{82}{81}$
------	------	--------------------	--------------------